

# MADISON COOTS

Harvard University · Cambridge, MA 02138

mcoots@g.harvard.edu | (510) 219-1084

madisoncoots.com

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## EDUCATION

### Harvard Kennedy School

*Ph.D. Public Policy, 2022–2027 (expected)*

### Harvard University

*M.A. Public Policy, 2022–2025*

### Stanford University

*M.S. Computer Science (Artificial Intelligence), 2019–2021*

### Stanford University

*B.S. Management Science and Engineering (Minor: English), 2015–2019*

## RESEARCH INTERESTS

Responsible AI, inequality, and the use of computational methods to advance organizational and policy goals

## REFERENCES

Sharad Goel

*Professor of Public Policy*

Harvard Kennedy School

[sgoel@hks.harvard.edu](mailto:sgoel@hks.harvard.edu)

Julian Nyarko

*Professor of Law*

Stanford Law School

[jnyarko@law.stanford.edu](mailto:jnyarko@law.stanford.edu)

Ravi Shroff

*Associate Professor of Applied Statistics*

New York University

[ravi.shroff@nyu.edu](mailto:ravi.shroff@nyu.edu)

## JOB MARKET PAPER

### Algorithmic Bias in Lending: Evidence from a Fintech Audit PREPRINT

Madison Coots, Robert Bartlett, Julian Nyarko, and Sharad Goel

*Under review*

Algorithmic lending has transformed the consumer credit landscape, with machine learning models commonly facilitating underwriting decisions. To comply with fair lending laws, these algorithms exclude legally protected characteristics, such as race and gender. Yet algorithmic underwriting can still inadvertently favor certain groups, prompting concerns about whether lending algorithms exhibit discriminatory behavior. Using proprietary loan-level data from a major U.S. fintech platform, we audit lending decisions across approximately 80,000 personal loans. We find that loans made to men and Black borrowers yielded lower profits than loans to other groups, suggesting that men and Black borrowers benefited from relatively favorable pricing. We trace these disparities to miscalibration in the platform’s underwriting model, which overestimates risk for women and underestimates risk for Black borrowers. We then show that one could correct this miscalibration—and the corresponding disparities—by including race and gender in underwriting models, illustrating a tension between competing notions of fairness.

## PEER-REVIEWED PUBLICATIONS

1. Alex Chohlas-Wood, **Madison Coots**, Joe Nudell, Julian Nyarko, Emma Brunskill, Todd Rogers, and Sharad Goel. “Automated Court Date Reminders Reduce Warrants for Arrest: Evidence from a Text Messaging Experiment.” *Science Advances* (2025). [PAPER](#)
2. **Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. “Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.” *Annual Review of Public Health* (2025). [PAPER](#)
3. **Madison Coots**, Soroush Saghaian, David Kent, and Sharad Goel. “A Framework for Considering the Role of Race and Ethnicity in Estimating Disease Risk.” *Annals of Internal Medicine* (2024). [PAPER](#)
4. Alex Chohlas-Wood, **Madison Coots**, Henry Zhu, Sharad Goel, and Emma Brunskill. “Learning to be Fair: A Consequentialist Approach to Equitable Decision-Making.” *Management Science* (2024). [PAPER](#)
5. Alex Chohlas-Wood\*, **Madison Coots\***, Julian Nyarko\*, and Sharad Goel\*. “Designing Equitable Algorithms.” *Nature Computational Science*, Vol. 3 (2023). [PAPER](#) \*Authors contributed equally.

## BOOK CHAPTERS

6. **Madison Coots**, Kristin A. Linn, Sharad Goel, Amol S. Navathe, and Ravi B. Parikh. “Racial Bias in Clinical and Population Health Algorithms: A Critical Review of Current Debates.” Chapter in *Social Stratification*, 5th Edition, edited by Michelle Jackson, Nima Dahir, Claire Daviss, and David B. Grusky. Routledge (2027).

## RESEARCH IN PROGRESS

7. **Madison Coots**, Hans Gaebler, Josh Grossman, Alex Chohlas-Wood, Julian Nyarko, Ravi Shroff, and Sharad Goel. “Towards Individualized Risk Predictions with LLMs.” In preparation (2026).
8. **Madison Coots**, Calvin Isley, Elisa Xi Chen, Nick Masi, Joe Nudell, Charlotte Tuminelli, Teddy Svoronos, and Sharad Goel. “AI Tutoring and Teacher Support Tools for Elementary Education.” In preparation (2026).

## HONORS AND AWARDS

|                                                 |                                                                |              |
|-------------------------------------------------|----------------------------------------------------------------|--------------|
| Rising Star in Data Science                     | <i>Univ. of Chicago, Harvard, Stanford, &amp; UC San Diego</i> | 2026         |
| Distinction in Student Teaching Award           | <i>Harvard Kennedy School</i>                                  | 2026         |
| Dissertation Research Grant (\$10,000)          | <i>Horowitz Foundation for Social Policy</i>                   | 2026         |
| Rising Star in IOE-ISyE-MS&E                    | <i>Univ. of Michigan, Georgia Tech, &amp; Stanford</i>         | 2026         |
| Dean’s Award for Excellence in Student Teaching | <i>Harvard Kennedy School</i>                                  | 2025         |
| Distinction in Student Teaching Award           | <i>Harvard Kennedy School</i>                                  | 2025         |
| Social Equity and Health Equity Grant (\$5,000) | <i>Malcolm Wiener Center, Harvard Kennedy School</i>           | 2025         |
| James M. and Cathleen D. Stone PhD Scholar      | <i>Harvard University</i>                                      | 2023–present |
| Harvard Graduate Prize Fellowship               | <i>Harvard University</i>                                      | 2022–2023    |
| Stanford Engineering Coterminial Fellowship     | <i>Stanford University</i>                                     | 2019–2020    |
| Stokes Scholar                                  | <i>U.S. Federal Government</i>                                 | 2017–2021    |

## INVITED TALKS

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|-----------------------------------------------------------------------------------|------|
| Harvard Applied Statistics Workshop (forthcoming) . . . . .                       | 2026 |
| Harvard Kennedy School Data + Donuts Series (forthcoming) . . . . .               | 2026 |
| IOE-ISyE-MS&E Rising Stars Workshop (Univ. of Michigan) . . . . .                 | 2026 |
| Predictive Analytics & Comparative Effectiveness Center Symposium (Tufts Univ.) . | 2024 |
| Guest Lecture in MS&E 298: Detecting Discrimination with Data (Stanford Univ.) .  | 2023 |
| Internet Institute Speaker Series (Oxford Univ.) . . . . .                        | 2021 |

## CONFERENCE PRESENTATIONS

**2026:** INFORMS Annual Meeting (forthcoming; contributed); APPAM Fall Research Conference (forthcoming; contributed); Conference on Empirical Legal Studies (contributed); International Conference on Computational Social Science (contributed); Econometric Society ESIF Economics and AI+ML Conference (contributed); POMS Annual Conference (contributed)

**2025:** American Law and Economics Association Conference (contributed); POMS Annual Conference (invited)

**2024:** Society for Medical Decision Making Annual Meeting (contributed); International Conference on Computational Social Science (contributed)

**2023:** Computational and Methodological Statistics Conference (invited); APPAM Fall Research Conference (contributed); INFORMS Annual Meeting (invited); INFORMS Healthcare Conference (invited)

**2022:** ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (contributed)

## TEACHING

### **API 201: Quantitative Analysis and Empirical Methods I**

*Teaching Fellow, Harvard Kennedy School, Fall 2024, 2025*

Applied statistics course in the MPP core: exploring and summarizing data with R, probability theory, statistical inference, and decision analysis

### **API 202Z: Quantitative Analysis and Empirical Methods II**

*Teaching Fellow, Harvard Kennedy School, Spring 2025, 2026*

Advanced applied statistics course in the MPP core: statistical inference and linear regression models

### **API 203Z: Quantitative Analysis and Empirical Methods III**

*Teaching Fellow, Harvard Kennedy School, Spring 2025*

Advanced applied statistics course in the MPP core: causal inference and machine learning

### **DPI 681M: The Science and Implications of Generative AI**

*Teaching Fellow, Harvard Kennedy School, Spring 2024*

Designed and taught the technical complement on language models, deep learning models, and transformers

### **MS&E 125: Applied Statistics**

*Course Assistant, Stanford University, School of Engineering, Winter 2020*

Undergraduate course in applied statistics: exploring and summarizing data, statistical inference, linear and logistic regression

## MS&E 252: Foundations of Decision Analysis

*Course Assistant, Stanford University, School of Engineering, Fall 2019*

Graduate course in quantitative decision analysis covering utility theory, sensitivity analysis, and value of information

## Code in Place

*Volunteer Section Leader, Stanford University, Spring 2020*

A 5-week introductory Python course offered during COVID-19 with 10,000 global students and 900 volunteer teachers; prepared and taught a weekly discussion section of 10–12 students

## PROFESSIONAL EXPERIENCE

### Systems & Technology Research

Senior Data Scientist (Part-time), Woburn, MA, 2023–present

### Stanford Computational Policy Lab

Data Scientist, Stanford, CA, 2020–2022

### Aerospace Technical Services

Data Scientist (Part-time), Remote, 2020–2024

### Central Intelligence Agency

Data Science Intern, Washington, D.C., 2017–2021

## ACADEMIC SERVICE

### Refereeing

Epidemiology..... 2025

International Conference on Computational Social Science..... 2025, 2026

### Leadership

Session Chair, Generative AI and Societal Sustainable Operations Track

*Production and Operations Management Society (POMS) Conference* ..... 2027

Organizing Team Member, Teaching Statistics in the Age of Agentic AI Workshop

*Harvard Kennedy School*..... 2026

Supporting Instructor, An Introduction to Reproducible Programming and Project Management Short Course *Society of Medical Decision Making Annual Meeting*..... 2024

## SKILLS

**Languages:** R, Python

**AI:** LLM evaluation, text data processing, chatbot design and user experience for AI systems, agentic workflows

**Methods:** Statistical inference, causal inference, simulation, decision analysis

## ADDITIONAL INFORMATION

### Hobbies & Interests

Plant cultivation and collection (specializing in Hoya species), bird watching, baking

### Citizenship

United States, Mexico